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MpSub: A Momentum p-Dimensional Subspace Trust-Region Method for Derivative-Free Fine-Tuning of Large Language Models

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Abstract Zeroth-order optimization avoids the activation memory of backpropagation when fine-tuning large language models, but existing methods rely on prescribed learning rates that require costly per-task tuning. We propose `MpSub`, a momentum p-dimensional subspace trust-region method. At each iteration, `MpSub` builds a linear trust-region model via central differences over a p-dimensional subspace spanned by the last accepted displacement (momentum) and $p-1$ Gaussian directions. A single trust-region radius adapts via the agreement ratio to govern both the finite-difference perturbation scale and the step norm, replacing a manual learning rate with self-adjusting step-size control. Exploiting the near-orthogonality of high-dimensional Gaussian vectors, `MpSub` omits basis orthogonalization and regenerates directions in place from seeds in $\mathcal{O}(n)$ auxiliary memory. For smooth deterministic objectives, we analyze the non-orthogonal Gaussian frame, bound the finite-difference error, quantify the subspace gradient capture, and prove almost-sure global convergence ($\lim_{k\to\infty} \|\nabla f(\boldsymbol{x}_k)\|_2 = 0$) under a safeguarded radius update. Numerical experiments on a nonquadratic benchmark and language model fine-tuning tasks confirm the contribution of each algorithmic component and show that `MpSub` with a single default configuration matches the accuracy of grid-searched zeroth-order baselines while remaining robust across a wide range of initial radii.

Keywords Derivative-free optimization · Subspace trust region · Zeroth-order optimization · Large language model fine-tuning

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1 Introduction

Full-parameter fine-tuning of pretrained large language models (LLMs) [31] with first-order optimizers requires storing intermediate activations for backpropagation as well as optimizer states such as momentum and second-moment buffers, resulting in a memory footprint several times larger than inference. Although parameter-efficient fine-tuning (PEFT) methods such as LoRA [15] reduce the number of trainable weights, they still rely on the backward pass and its activation graph. Zeroth-order (ZO) optimization removes backpropagation altogether by estimating descent directions from forward evaluations alone, building on classical simultaneous perturbation stochastic approximation (SPSA) [24] and Gaussian randomized smoothing [10,12,22]. In particular, MeZO [20] adapts the two-point Gaussian estimator to LLM fine-tuning by regenerating parameter-sized perturbations in place from scalar random seeds, reducing auxiliary memory to the inference level. Subsequent work has extended this paradigm through low-dimensional subspace perturbations [28], low-rank gradient structures [6], sparse perturbation masks [19], and first- or second-order moment scaling [5,32].

Despite these advances, existing ZO fine-tuning methods remain within the open-loop stochastic gradient framework and share two structural limitations. First, the step size is prescribed by an open-loop learning-rate schedule rather than adjusted from observed objective reduction, and is chosen separately from the finite-difference perturbation radius. Because trial steps are applied unconditionally without verifying objective decrease, an overly large learning rate can push the iterate outside the neighborhood where the finite-difference approximation is accurate and cause divergence, whereas a small learning rate causes slow progress, necessitating costly multi-run grid searches. Second, sampling a single isotropic random direction in $\mathbb{R}^n$ captures only an $\mathcal{O}(1/n)$ fraction of the squared gradient norm in expectation and retains no memory of previous successful steps; incorporating full-space first- or second-moment buffers [5,32] restores history across iterations, but requires storing additional n-dimensional state vectors.

In mathematical optimization, derivative-free trust-region (DFO-TR) algorithms overcome step-size sensitivity through closed-loop radius control [1,7, 9,8,18,23]. At each iteration, a DFO-TR method constructs a local surrogate model within a trust region of radius Δ_k, evaluates the trial step via the agreement ratio between actual and predicted reductions, and expands or contracts Δ_k accordingly. Because building a full-space linear or quadratic interpolation model requires $\mathcal{O}(n)$ or $\mathcal{O}(n^2)$ function evaluations per step, large-scale DFO methods restrict model construction to a low-dimensional subspace at each iteration [26,29,30]. Recent randomized subspace DFO methods sample low-dimensional subspaces via random projections or Gaussian matrices, analyzing finite-difference descent [16,17], Johnson–Lindenstrauss subspace trust-region models [4,3,11], and linear model accuracy over random subspaces [14,21]. Closest to our approach, 2D-MoSub [27] spans a two-dimensional subspace at each step using the previous accepted displacement (momentum) and one

fresh random direction, minimizing a quadratic interpolation model within the subspace trust region to combine trajectory reuse with random exploration.

At LLM scale ($n \sim 10^8$), however, pairing momentum with only a single random direction ($p = 2$) explores too little of the ambient space whenever the momentum direction ceases to yield descent, motivating a larger subspace dimension $p > 2$. Yet scaling subspace DFO-TR methods to p-dimensional subspaces at this scale encounters three obstacles. First, because the exploratory directions are redrawn at every step, the local model must be constructed from scratch on each new subspace; quadratic interpolation in p dimensions requires $\frac{1}{2}(p+1)(p+2)$ function evaluations per iteration (231 forward passes at $p = 20$), which is prohibitive when each evaluation is a language model forward pass. Second, existing randomized subspace DFO methods [4,3,16] explicitly instantiate or orthogonalize an $n \times p$ basis matrix $\boldsymbol{D}_k$ (requiring $\mathcal{O}(pn)$ memory and $\mathcal{O}(p^2 n)$ orthogonalization operations), which at $p = 20$ would forfeit the $\mathcal{O}(n)$ memory advantage of zeroth-order fine-tuning. Third, when explicit orthogonalization is omitted to avoid storing $\boldsymbol{D}_k$ and a deterministic momentum direction is combined with $p - 1$ independent Gaussian vectors, the subspace columns form a hybrid non-orthogonal frame rather than an orthonormal or Haar-distributed basis, placing the method outside standard subspace trust-region convergence frameworks.

To address these challenges, we propose `MpSub`, a momentum p-dimensional subspace trust-region method designed for high-dimensional derivative-free optimization and LLM fine-tuning. At each iteration, `MpSub` constructs a p-dimensional subspace spanned by the normalized displacement of the most recent accepted step and $p - 1$ independent Gaussian directions, estimates the p directional derivatives via central finite differences, and computes the trial step from the resulting linear trust-region model. Replacing quadratic interpolation with a central-difference linear model reduces the per-iteration evaluation cost from $\mathcal{O}(p^2)$ to $2p + 2$ while coupling the finite-difference perturbation scale and the step norm to a single adaptive trust-region radius Δ_k. Furthermore, because independent Gaussian vectors in $\mathbb{R}^n$ have root-mean-square inner product $n^{-1/2}$ (10^{-4} at $n \sim 10^8$), `MpSub` dispenses with Gram–Schmidt orthogonalization and regenerates the $p-1$ exploratory directions in place from scalar random seeds, maintaining $\mathcal{O}(n)$ auxiliary memory alongside a single stored momentum buffer.

For smooth deterministic objectives, we analyze the non-orthogonal Gaussian frame directly without imposing column orthogonality: we bound the central-difference model error, prove that the subspace captures at least a $(p-1)/n$ fraction of the squared gradient norm in conditional expectation, and establish unconditional almost-sure global convergence ($\lim_{k\to\infty} \|\nabla f(\boldsymbol{x}_k)\|_2 = 0$) under a safeguarded radius update. Numerically, component ablations on a 1,000-dimensional smooth nonquadratic benchmark verify the individual contributions of the momentum direction, subspace dimension p, decoupled strict-decrease acceptance rule, and adaptive radius control. On SuperGLUE CommitmentBank [25] fine-tuning of OPT-125M and OPT-350M under a matched budget of 8,400 forward passes, `MpSub` with a single default configuration

achieves mean test accuracies of 0.673 and 0.690, matching MeZO tuned via learning-rate grid search (0.685 on both models) while remaining robust to initial radius variations across two orders of magnitude.

The remainder of this paper is organized as follows. Section 2 formulates the `MpSub` algorithm and its seed-based $\mathcal{O}(n)$-memory implementation. Section 3 presents the deterministic convergence analysis. Section 4 reports the numerical experiments and ablations, and Section 5 concludes the paper.

2 The MpSub Algorithm

We consider the unconstrained derivative-free optimization problem

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \; f(\boldsymbol{x}),$$

where the parameter dimension n is large (e.g., $n \sim 10^8$ in full-parameter language model fine-tuning) and only function values are available. At each iteration k, `MpSub` maintains the current iterate $\boldsymbol{x}_k \in \mathbb{R}^n$, a trust-region radius $\Delta_k \in [\Delta_{\min}, \Delta_{\max}]$, and the displacement $\boldsymbol{m}_k \in \mathbb{R}^n$ from the most recent accepted step (initialized at $\boldsymbol{m}_0 = \boldsymbol{0}$).

2.1 Subspace trust-region iteration

At iteration k, `MpSub` restricts its search to the p-dimensional affine subspace $\boldsymbol{x}_k + \mathcal{S}_k$, where $\mathcal{S}_k = \operatorname{span}\{\boldsymbol{d}_1^{(k)}, \ldots, \boldsymbol{d}_p^{(k)}\} \subset \mathbb{R}^n$ ($p \geq 2$). Collecting the p search directions as columns of the frame matrix $\boldsymbol{D}_k = [\boldsymbol{d}_1^{(k)}, \ldots, \boldsymbol{d}_p^{(k)}] \in \mathbb{R}^{n\times p}$, any candidate point in $\boldsymbol{x}_k + \mathcal{S}_k$ is parameterized by a subspace coordinate vector $\boldsymbol{s}_s \in \mathbb{R}^p$ via

$$\boldsymbol{x}(\boldsymbol{s}_s) = \boldsymbol{x}_k + \boldsymbol{D}_k \boldsymbol{s}_s = \boldsymbol{x}_k + \sum_{i=1}^{p} s_{s,i} \boldsymbol{d}_i^{(k)}.$$

Whenever a prior accepted step is available ($\|\boldsymbol{m}_k\|_2 > 0$), the first direction reuses the normalized momentum displacement

$$\boldsymbol{d}_1^{(k)} = \frac{\boldsymbol{m}_k}{\|\boldsymbol{m}_k\|_2}, \tag{1}$$

which embeds the most recent successful displacement into $\mathcal{S}_k$ at zero additional evaluation cost. The remaining $p-1$ exploratory directions are drawn independently as scaled Gaussian vectors,

$$\boldsymbol{d}_i^{(k)} = \frac{\boldsymbol{z}_i}{\sqrt{n}}, \qquad \boldsymbol{z}_i \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n), \quad i = 2, \ldots, p, \tag{2}$$

with the same distribution supplying $\boldsymbol{d}_1^{(k)}$ when $\boldsymbol{m}_k = \boldsymbol{0}$.

In classical subspace trust-region methods, the columns of $\boldsymbol{D}_k$ are orthonormalized ($\boldsymbol{D}_k^\top \boldsymbol{D}_k = \boldsymbol{I}_p$) so that the subspace ball $\|\boldsymbol{s}_s\|_2 \leq \Delta_k$ maps isometrically into $\mathbb{R}^n$. Applying Gram–Schmidt to $\boldsymbol{d}_1^{(k)}, \ldots, \boldsymbol{d}_p^{(k)}$, however, requires

storing all p vectors in $\mathbb{R}^n$ simultaneously ($\mathcal{O}(pn)$ memory) and performing $\mathcal{O}(p^2 n)$ arithmetic, while coupling each column to all preceding columns so that directions can no longer be regenerated individually from scalar seeds. The following proposition shows that when $n \gg p$, the unorthogonalized hybrid frame $\boldsymbol{D}_k$ defined by (1)–(2) is already an approximate isometry in conditional expectation.

Proposition 1 (Near-orthonormality of the hybrid frame) *Let $\mathcal{F}_k$ denote the σ-algebra generated by the random directions drawn prior to iteration k. For the frame $\boldsymbol{D}_k = [\boldsymbol{d}_1^{(k)}, \dots, \boldsymbol{d}_p^{(k)}]$ constructed via (1)–(2), it holds for all $1 \le i \le p$ and $1 \le i < j \le p$ that*

$$\mathbb{E}\big[\|\boldsymbol{d}_i^{(k)}\|_2^2 \,\big|\, \mathcal{F}_k\big] = 1, \quad \mathbb{E}\big[\boldsymbol{d}_i^{(k)\top} \boldsymbol{d}_j^{(k)} \,\big|\, \mathcal{F}_k\big] = 0, \quad \mathbb{E}\big[\big(\boldsymbol{d}_i^{(k)\top} \boldsymbol{d}_j^{(k)}\big)^2 \,\big|\, \mathcal{F}_k\big] = \frac{1}{n}. \tag{3}$$

Consequently, the Gram matrix $\boldsymbol{D}_k^\top \boldsymbol{D}_k$ satisfies $\mathbb{E}[\boldsymbol{D}_k^\top \boldsymbol{D}_k \mid \mathcal{F}_k] = \boldsymbol{I}_p$ and

$$\mathbb{E}\big[\|\boldsymbol{D}_k^\top \boldsymbol{D}_k - \boldsymbol{I}_p\|_F^2 \,\big|\, \mathcal{F}_k\big] \le \frac{p(p+1)}{n}.$$

Proof Conditioned on $\mathcal{F}_k$, if $\|\boldsymbol{m}_k\|_2 > 0$, then $\boldsymbol{d}_1^{(k)} = \boldsymbol{m}_k / \|\boldsymbol{m}_k\|_2$ is $\mathcal{F}_k$-measurable with $\|\boldsymbol{d}_1^{(k)}\|_2^2 = 1$; if $\boldsymbol{m}_k = \boldsymbol{0}$, then $\boldsymbol{d}_1^{(k)} = \boldsymbol{z}_1/\sqrt{n}$ with $\boldsymbol{z}_1 \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n)$ independent of $\mathcal{F}_k$, so $\mathbb{E}[\|\boldsymbol{d}_1^{(k)}\|_2^2 \mid \mathcal{F}_k] = \frac{1}{n}\mathbb{E}[\|\boldsymbol{z}_1\|_2^2] = 1$. The same identity gives $\mathbb{E}[\|\boldsymbol{d}_i^{(k)}\|_2^2 \mid \mathcal{F}_k] = 1$ for $i = 2, \dots, p$.

For $1 \le i < j \le p$, since $j \ge 2$, $\boldsymbol{d}_j^{(k)} = \boldsymbol{z}_j/\sqrt{n}$ is independent of both $\mathcal{F}_k$ and $\boldsymbol{d}_i^{(k)}$, with $\mathbb{E}[\boldsymbol{d}_j^{(k)} \mid \mathcal{F}_k, \boldsymbol{d}_i^{(k)}] = \boldsymbol{0}$ and $\mathbb{E}[\boldsymbol{d}_j^{(k)} \boldsymbol{d}_j^{(k)\top} \mid \mathcal{F}_k, \boldsymbol{d}_i^{(k)}] = \frac{1}{n}\boldsymbol{I}_n$. By the tower property,

$$\begin{aligned}
\mathbb{E}\big[\boldsymbol{d}_i^{(k)\top} \boldsymbol{d}_j^{(k)} \,\big|\, \mathcal{F}_k\big] &= \mathbb{E}\Big[\boldsymbol{d}_i^{(k)\top} \mathbb{E}\big[\boldsymbol{d}_j^{(k)} \,\big|\, \mathcal{F}_k, \boldsymbol{d}_i^{(k)}\big] \,\Big|\, \mathcal{F}_k\Big] = 0, \\
\mathbb{E}\big[\big(\boldsymbol{d}_i^{(k)\top} \boldsymbol{d}_j^{(k)}\big)^2 \,\big|\, \mathcal{F}_k\big] &= \mathbb{E}\Big[\boldsymbol{d}_i^{(k)\top} \mathbb{E}\big[\boldsymbol{d}_j^{(k)} \boldsymbol{d}_j^{(k)\top} \,\big|\, \mathcal{F}_k, \boldsymbol{d}_i^{(k)}\big] \boldsymbol{d}_i^{(k)} \,\Big|\, \mathcal{F}_k\Big] \\
&= \frac{1}{n}\mathbb{E}\big[\|\boldsymbol{d}_i^{(k)}\|_2^2 \,\big|\, \mathcal{F}_k\big] = \frac{1}{n},
\end{aligned}$$

which proves (3) and $\mathbb{E}[\boldsymbol{D}_k^\top \boldsymbol{D}_k \mid \mathcal{F}_k] = \boldsymbol{I}_p$. Finally, for each Gaussian column $\boldsymbol{d}_i^{(k)} = \boldsymbol{z}_i/\sqrt{n}$, $\|\boldsymbol{z}_i\|_2^2 \sim \chi_n^2$ has variance $2n$, so $\mathbb{E}[(\|\boldsymbol{d}_i^{(k)}\|_2^2 - 1)^2 \mid \mathcal{F}_k] = 2/n$, whereas if $\|\boldsymbol{m}_k\|_2 > 0$, $(\|\boldsymbol{d}_1^{(k)}\|_2^2 - 1)^2 = 0 \le 2/n$. Summing the p diagonal terms and the $p(p-1)$ off-diagonal terms of $\boldsymbol{D}_k^\top \boldsymbol{D}_k - \boldsymbol{I}_p$ yields $\mathbb{E}[\|\boldsymbol{D}_k^\top \boldsymbol{D}_k - \boldsymbol{I}_p\|_F^2 \mid \mathcal{F}_k] \le \frac{2p + p(p-1)}{n} = \frac{p(p+1)}{n}$. □

At $n \sim 10^8$ and $p = 20$, Proposition 1 implies a root-mean-square Frobenius deviation $(\mathbb{E}[\|\boldsymbol{D}_k^\top \boldsymbol{D}_k - \boldsymbol{I}_p\|_F^2 \mid \mathcal{F}_k])^{1/2} \le \sqrt{p(p+1)/n} \approx 2 \times 10^{-3}$, justifying the direct use of $\boldsymbol{D}_k$ without Gram–Schmidt orthogonalization.

To construct a local model over $\mathcal{S}_k$ with $O(p)$ rather than $O(p^2)$ function evaluations, `MpSub` estimates the directional derivative along each $\boldsymbol{d}_i^{(k)}$ via

central finite differences at perturbation scale Δ_k:

$$g_{k,i} = \frac{f(\boldsymbol{x}_k + \Delta_k \boldsymbol{d}_i^{(k)}) - f(\boldsymbol{x}_k - \Delta_k \boldsymbol{d}_i^{(k)})}{2\Delta_k}, \quad i = 1, \dots, p. \tag{4}$$

Using Δ_k as the finite-difference step couples the derivative sampling stencil to the radius of the linear surrogate model

$$m_k(\boldsymbol{s}_s) = f(\boldsymbol{x}_k) + \boldsymbol{g}_k^\top \boldsymbol{s}_s, \tag{5}$$

where $\boldsymbol{g}_k = (g_{k,1}, \dots, g_{k,p})^\top \in \mathbb{R}^p$ approximates the exact subspace gradient $\boldsymbol{D}_k^\top \nabla f(\boldsymbol{x}_k)$ with an $O(\Delta_k)$ error under Lipschitz continuity of ∇f (Lemma 1). The trial coordinate step $\boldsymbol{s}_s$ is obtained by solving the Euclidean trust-region subproblem

$$\min_{\boldsymbol{s}_s \in \mathbb{R}^p} \ \boldsymbol{g}_k^\top \boldsymbol{s}_s \quad \text{subject to} \quad \|\boldsymbol{s}_s\|_2 \le \Delta_k.$$

By the Cauchy–Schwarz inequality, $\boldsymbol{g}_k^\top \boldsymbol{s}_s \ge -\|\boldsymbol{g}_k\|_2 \|\boldsymbol{s}_s\|_2 \ge -\Delta_k \|\boldsymbol{g}_k\|_2$ for all $\|\boldsymbol{s}_s\|_2 \le \Delta_k$; whenever $\boldsymbol{g}_k \neq \boldsymbol{0}$, equality is attained uniquely at the closed-form minimizer

$$\boldsymbol{s}_s = -\Delta_k \frac{\boldsymbol{g}_k}{\|\boldsymbol{g}_k\|_2}, \tag{6}$$

which yields the ambient trial iterate and predicted model reduction

$$\begin{aligned} \boldsymbol{x}^+ &= \boldsymbol{x}_k + \boldsymbol{D}_k \boldsymbol{s}_s = \boldsymbol{x}_k - \frac{\Delta_k}{\|\boldsymbol{g}_k\|_2} \sum_{i=1}^{p} g_{k,i} \boldsymbol{d}_i^{(k)}, \\ \mathrm{pred}_k &= m_k(\boldsymbol{0}) - m_k(\boldsymbol{s}_s) = \Delta_k \|\boldsymbol{g}_k\|_2 > 0. \end{aligned} \tag{7}$$

In the degenerate event $\boldsymbol{g}_k = \boldsymbol{0}$, the algorithm sets $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k$, $\boldsymbol{m}_{k+1} = \boldsymbol{m}_k$, and $\Delta_{k+1} = \max\{\gamma_1 \Delta_k, \Delta_{\min}\}$ without evaluating $\boldsymbol{x}^+$.

After evaluating $f(\boldsymbol{x}^+)$, the agreement between the linear model (5) and the objective is measured by the ratio

$$\rho_k = \frac{f(\boldsymbol{x}_k) - f(\boldsymbol{x}^+)}{\Delta_k \|\boldsymbol{g}_k\|_2}. \tag{8}$$

In classical trust-region methods [7], the condition $\rho_k \ge \eta$ governs both iterate acceptance and radius expansion. In derivative-free subspace optimization, however, $2p + 1$ function evaluations have already been expended to form $\boldsymbol{g}_k$ before $f(\boldsymbol{x}^+)$ is tested. Rejecting a trial step that achieves actual descent ($f(\boldsymbol{x}^+) < f(\boldsymbol{x}_k)$) when $0 < \rho_k < \eta$ would discard that objective reduction and leave the momentum displacement $\boldsymbol{m}_k$ unchanged. `MpSub` therefore decouples step acceptance from radius control: any trial iterate that strictly reduces the objective is accepted and stored as the new momentum displacement,

$$(\boldsymbol{x}_{k+1}, \boldsymbol{m}_{k+1}) = \begin{cases} (\boldsymbol{x}^+,\ \boldsymbol{x}^+ - \boldsymbol{x}_k), & \text{if } f(\boldsymbol{x}^+) < f(\boldsymbol{x}_k), \\ (\boldsymbol{x}_k,\ \boldsymbol{m}_k), & \text{otherwise,} \end{cases} \tag{9}$$

while the agreement ratio ρ_k independently updates the trust-region radius via $\eta \in (0,1)$ and $0 < \gamma_1 < 1 < \gamma_2$:

$$\Delta_{k+1} = \begin{cases} \min\{\gamma_2 \Delta_k, \Delta_{\max}\}, & \text{if } \rho_k \geq \eta, \\ \max\{\gamma_1 \Delta_k, \Delta_{\min}\}, & \text{if } \rho_k < \eta. \end{cases} \tag{10}$$

When $\rho_k \geq \eta$, Δ_k expands to allow larger steps; when $\rho_k < \eta$, Δ_k contracts, simultaneously shortening the next trial step and refining the central-difference stencil (4). The complete iteration is summarized in Algorithm 1.

Algorithm 1 `MpSub`

Input: Initial iterate $\boldsymbol{x}_0 \in \mathbb{R}^n$, subspace dimension $2 \leq p \leq n$, initial radius $\Delta_0 \in [\Delta_{\min}, \Delta_{\max}]$, factors $0 < \gamma_1 < 1 < \gamma_2$, bounds $0 < \Delta_{\min} < \Delta_{\max}$, and threshold $\eta \in (0,1)$.
Initialize momentum displacement $\boldsymbol{m}_0 = \boldsymbol{0}$.
for $k = 0, 1, 2, \ldots$ **do**
  1. Subspace generation: Set $\boldsymbol{d}_1^{(k)}$ via (1) if $\|\boldsymbol{m}_k\|_2 > 0$, or draw $\boldsymbol{d}_1^{(k)} = \boldsymbol{z}_1/\sqrt{n}$ with $\boldsymbol{z}_1 \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n)$ otherwise; draw $\boldsymbol{d}_i^{(k)} = \boldsymbol{z}_i/\sqrt{n}$ with $\boldsymbol{z}_i \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_n)$ independently for $i = 2, \ldots, p$.
  2. Subspace gradient estimation: Evaluate $g_{k,i}$ for $i = 1, \ldots, p$ via (4) and set $\boldsymbol{g}_k = (g_{k,1}, \ldots, g_{k,p})^\top$.
  3. Trust-region trial step: If $\boldsymbol{g}_k = \boldsymbol{0}$, set $(\boldsymbol{x}_{k+1}, \boldsymbol{m}_{k+1}) = (\boldsymbol{x}_k, \boldsymbol{m}_k)$ and $\Delta_{k+1} = \max\{\gamma_1 \Delta_k, \Delta_{\min}\}$, and proceed to $k+1$. Otherwise, compute $\boldsymbol{s}_s = -\Delta_k \boldsymbol{g}_k / \|\boldsymbol{g}_k\|_2$, trial iterate $\boldsymbol{x}^+$ via (7), and agreement ratio ρ_k via (8).
  4. Acceptance and radius update: Update $(\boldsymbol{x}_{k+1}, \boldsymbol{m}_{k+1})$ via (9) and Δ_{k+1} via (10).
end for

2.2 Seed-based realization for stochastic LLM fine-tuning

In full-parameter LLM fine-tuning, the training loss is evaluated on stochastic minibatches and the parameter dimension n precludes storing p direction vectors. The structure of Algorithm 1 allows both issues to be addressed without altering the mathematical update.

First, at iteration k, let $\mathcal{B}_k \subset \mathcal{D}$ denote the sampled minibatch and define the empirical minibatch objective

$$f_k(\boldsymbol{x}) = \frac{1}{|\mathcal{B}_k|} \sum_{\xi \in \mathcal{B}_k} \ell(\boldsymbol{x}; \xi).$$

If different minibatches were drawn within the same iteration, the finite differences (4) and the numerator $f_k(\boldsymbol{x}_k) - f_k(\boldsymbol{x}^+)$ of ρ_k would mix evaluations of distinct sample functions, confounding model agreement with minibatch sampling variance. `MpSub` therefore holds the minibatch $\mathcal{B}_k$ fixed across all $2p+2$ forward evaluations of iteration k—namely $f_k(\boldsymbol{x}_k)$, the $2p$ perturbed evaluations $f_k(\boldsymbol{x}_k \pm \Delta_k \boldsymbol{d}_i^{(k)})$, and the trial evaluation $f_k(\boldsymbol{x}^+)$—so that iteration k

executes a well-defined trust-region step on the single empirical objective f_k, before drawing a fresh minibatch $\mathcal{B}_{k+1}$ at step $k+1$.

Second, explicitly materializing $\boldsymbol{D}_k \in \mathbb{R}^{n\times p}$ would require pn floating-point numbers (about 10 GB in FP32 for OPT-125M at $n \approx 1.25\times 10^8$ and $p = 20$, exceeding the model parameters themselves). Because Proposition 1 allows `MpSub` to omit Gram–Schmidt orthogonalization, each exploratory direction $\boldsymbol{d}_i^{(k)}$ $(i = 2,\dots,p)$ depends only on its own independent Gaussian draw and can be identified with a scalar pseudorandom seed $s_{k,i}$. The algorithm therefore stores only one $\mathcal{O}(n)$ auxiliary buffer for $\boldsymbol{m}_k$ and executes all perturbations in place on the model weights $\boldsymbol{x}_k$: during central-difference estimation, the weights are shifted by $\pm\Delta_k \boldsymbol{d}_1^{(k)}$ using $\boldsymbol{m}_k$ and by $\pm\Delta_k \boldsymbol{d}_i^{(k)}$ $(i \geq 2)$ by resetting the pseudorandom generator to seed $s_{k,i}$ for the positive perturbation, the negative perturbation, and the restoration to $\boldsymbol{x}_k$. Once $\boldsymbol{s}_s = -\Delta_k \boldsymbol{g}_k/\|\boldsymbol{g}_k\|_2$ is obtained, reseeding $\{s_{k,i}\}_{i=2}^{p}$ accumulates the trial displacement

$$\boldsymbol{u}_k = s_{s,1}\boldsymbol{d}_1^{(k)} + \sum_{i=2}^{p} s_{s,i}\boldsymbol{d}_i^{(k)}$$

directly into the weights to form $\boldsymbol{x}^+ = \boldsymbol{x}_k + \boldsymbol{u}_k$. If $f_k(\boldsymbol{x}^+) < f_k(\boldsymbol{x}_k)$, the weights remain at $\boldsymbol{x}^+$ and the same seeds overwrite the momentum buffer $\boldsymbol{m}_{k+1} = \boldsymbol{u}_k$ in place; if $f_k(\boldsymbol{x}^+) \geq f_k(\boldsymbol{x}_k)$, the seed loop subtracts $\boldsymbol{u}_k$ in place to restore $\boldsymbol{x}_k$ while leaving $\boldsymbol{m}_{k+1} = \boldsymbol{m}_k$ untouched. Consequently, peak auxiliary memory remains $\mathcal{O}(n)$—a single parameter-sized vector $\boldsymbol{m}_k$—regardless of the subspace dimension p.

3 Theoretical Analysis

We now establish global convergence of `MpSub` (Algorithm 1) on smooth deterministic objectives. Compared with classical subspace trust-region analyses [4,3], two structural features of `MpSub` must be accounted for: first, the hybrid frame $\boldsymbol{D}_k = [\boldsymbol{d}_1^{(k)},\dots,\boldsymbol{d}_p^{(k)}]$ defined by (1)–(2) is not orthogonalized and its Gaussian columns have random, unbounded norms; second, the subspace gradient $\boldsymbol{g}_k$ is estimated by central finite differences (4) at the trust-region scale Δ_k rather than computed from exact derivatives. Our analysis therefore proceeds in two stages. In Section 3.1, we first derive deterministic finite-difference error and single-step descent bounds for an arbitrary frame $\boldsymbol{D}_k$, and then use the conditional Gaussian law of the exploratory columns to show that $\boldsymbol{D}_k$ captures a non-vanishing fraction of $\|\nabla f(\boldsymbol{x}_k)\|_2$ while keeping the frame norms bounded with controllable conditional probability. In Section 3.2, we combine these single-step guarantees with a martingale drift argument and a conditional expectation transfer over large-gradient iterations to prove unconditional almost-sure convergence $\lim_{k\to\infty}\|\nabla f(\boldsymbol{x}_k)\|_2 = 0$.

Throughout this section, we set $\Delta_{\min} = 0$ to study the asymptotic regime $k \to \infty$, and write $\boldsymbol{x}_k^+$ for the trial point $\boldsymbol{x}^+$ at iteration k. Following standard trust-region convergence theory [9], we couple radius expansion with a

criticality safeguard for a fixed constant $\kappa > 0$:

$$\Delta_{k+1} = \begin{cases} \min\{\gamma_2 \Delta_k, \Delta_{\max}\}, & \text{if } \rho_k \geq \eta \text{ and } \|\boldsymbol{g}_k\|_2 \geq \kappa \Delta_k, \\ \gamma_1 \Delta_k, & \text{otherwise.} \end{cases} \tag{11}$$

As in (9), a trial point is accepted ($\boldsymbol{x}_{k+1} = \boldsymbol{x}_k^+$) if and only if $f(\boldsymbol{x}_k^+) < f(\boldsymbol{x}_k)$. The condition $\|\boldsymbol{g}_k\|_2 \geq \kappa \Delta_k$ in (11) prevents the trust-region radius from expanding when the estimated subspace gradient is already smaller than $\mathcal{O}(\Delta_k)$, ensuring that every expansion step yields an objective reduction of at least $\eta \kappa \Delta_k^2$; in finite-budget runs with $\Delta_{\min} > 0$, setting $\kappa = 0$ as in (10) produces virtually identical trajectories (Section 4).

Assumption 1 *The objective function $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and bounded below by $f_{\text{low}} > -\infty$, and its gradient ∇f is Lipschitz continuous with constant $L > 0$:*

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\|_2 \leq L \|\boldsymbol{x} - \boldsymbol{y}\|_2, \qquad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$$

3.1 Single-step descent

For a general frame $\boldsymbol{D}_k = [\boldsymbol{d}_1^{(k)}, \ldots, \boldsymbol{d}_p^{(k)}] \in \mathbb{R}^{n \times p}$ at iteration k, let

$$\hat{g}_{k,i} := \nabla f(\boldsymbol{x}_k)^\top \boldsymbol{d}_i^{(k)} \quad (i = 1, \ldots, p), \qquad \hat{\boldsymbol{g}}_k := \boldsymbol{D}_k^\top \nabla f(\boldsymbol{x}_k),$$

denote the exact directional derivatives and the exact subspace gradient, and let $\boldsymbol{g}_k = (g_{k,1}, \ldots, g_{k,p})^\top$ denote the central-difference estimator defined in (4). Because the columns $\boldsymbol{d}_i^{(k)}$ are not normalized to unit length, both the truncation error of $\boldsymbol{g}_k$ and the ambient displacement norm $\|\boldsymbol{D}_k \boldsymbol{s}_s\|_2$ depend explicitly on the column norms and spectral norm of $\boldsymbol{D}_k$.

Lemma 1 (Finite-difference error) *Suppose Assumption 1 holds. Then for any frame $\boldsymbol{D}_k = [\boldsymbol{d}_1^{(k)}, \ldots, \boldsymbol{d}_p^{(k)}]$,*

$$|g_{k,i} - \hat{g}_{k,i}| \leq \frac{L \Delta_k}{2} \|\boldsymbol{d}_i^{(k)}\|_2^2, \qquad i = 1, \ldots, p. \tag{12}$$

Consequently,

$$\|\boldsymbol{g}_k - \hat{\boldsymbol{g}}_k\|_2 \leq \frac{L \Delta_k}{2} Q_k, \qquad Q_k := \left(\sum_{i=1}^{p} \|\boldsymbol{d}_i^{(k)}\|_2^4 \right)^{1/2}. \tag{13}$$

Proof By Assumption 1, the standard descent inequality gives $|f(\boldsymbol{x}_k + t\boldsymbol{d}) - f(\boldsymbol{x}_k) - t \nabla f(\boldsymbol{x}_k)^\top \boldsymbol{d}| \leq \frac{1}{2} L t^2 \|\boldsymbol{d}\|_2^2$ for any $\boldsymbol{d} \in \mathbb{R}^n$ and $t > 0$. Applying this bound with $t = \Delta_k$ and $\boldsymbol{d} = \pm \boldsymbol{d}_i^{(k)}$ yields

$$f(\boldsymbol{x}_k \pm \Delta_k \boldsymbol{d}_i^{(k)}) = f(\boldsymbol{x}_k) \pm \Delta_k \hat{g}_{k,i} + R_{i,\pm}, \qquad |R_{i,\pm}| \leq \frac{L \Delta_k^2}{2} \|\boldsymbol{d}_i^{(k)}\|_2^2.$$

Subtracting the two expansions and dividing by $2\Delta_k$ gives $g_{k,i} - \hat{g}_{k,i} = (R_{i,+} - R_{i,-})/(2\Delta_k)$, which implies (12). Squaring and summing (12) over $i = 1, \ldots, p$ yields (13). □

Combining Lemma 1 with the closed-form solution (6) of the linear trust-region subproblem yields a deterministic lower bound on the objective decrease and agreement ratio ρ_k for any frame $\boldsymbol{D}_k$.

Proposition 2 (Single-step decrease for a general frame) *Suppose Assumption 1 holds and $\boldsymbol{g}_k \neq \boldsymbol{0}$. For the trial step $\boldsymbol{s}_s = -\Delta_k \boldsymbol{g}_k/\|\boldsymbol{g}_k\|_2$ and trial point $\boldsymbol{x}_k^+ = \boldsymbol{x}_k + \boldsymbol{D}_k \boldsymbol{s}_s$, let $\beta_k := \|\boldsymbol{D}_k\|_2$. Then*

$$f(\boldsymbol{x}_k) - f(\boldsymbol{x}_k^+) \geq \Delta_k \|\boldsymbol{g}_k\|_2 - \frac{L\Delta_k^2}{2}(Q_k + \beta_k^2), \tag{14}$$

and the agreement ratio (8) *satisfies*

$$\rho_k \geq 1 - \frac{L\Delta_k(Q_k + \beta_k^2)}{2\|\boldsymbol{g}_k\|_2}. \tag{15}$$

Proof Let $\boldsymbol{h}_k = \boldsymbol{D}_k \boldsymbol{s}_s$. Since $\hat{\boldsymbol{g}}_k = \boldsymbol{D}_k^\top \nabla f(\boldsymbol{x}_k)$ and $\boldsymbol{g}_k^\top \boldsymbol{s}_s = -\Delta_k \|\boldsymbol{g}_k\|_2$, Cauchy–Schwarz and Lemma 1 give

$$\begin{aligned}\nabla f(\boldsymbol{x}_k)^\top \boldsymbol{h}_k &= \boldsymbol{g}_k^\top \boldsymbol{s}_s + (\hat{\boldsymbol{g}}_k - \boldsymbol{g}_k)^\top \boldsymbol{s}_s \\ &\leq -\Delta_k \|\boldsymbol{g}_k\|_2 + \|\hat{\boldsymbol{g}}_k - \boldsymbol{g}_k\|_2 \|\boldsymbol{s}_s\|_2 \leq -\Delta_k \|\boldsymbol{g}_k\|_2 + \frac{LQ_k}{2}\Delta_k^2.\end{aligned}$$

Moreover, $\|\boldsymbol{h}_k\|_2 \leq \|\boldsymbol{D}_k\|_2 \|\boldsymbol{s}_s\|_2 = \beta_k \Delta_k$. Applying the L-smoothness upper bound $f(\boldsymbol{x}_k^+) \leq f(\boldsymbol{x}_k) + \nabla f(\boldsymbol{x}_k)^\top \boldsymbol{h}_k + \frac{L}{2}\|\boldsymbol{h}_k\|_2^2$ gives (14), and dividing by $\Delta_k \|\boldsymbol{g}_k\|_2$ yields (15). □

Proposition 2 shows that a trial step is guaranteed to achieve $\rho_k \geq \eta$ whenever Δ_k is small relative to $\|\boldsymbol{g}_k\|_2/(Q_k+\beta_k^2)$. To ensure that this condition holds with uniformly positive conditional probability when $\nabla f(\boldsymbol{x}_k) \neq \boldsymbol{0}$, we next characterize the gradient energy captured by the hybrid frame $\boldsymbol{D}_k$ defined in (1)–(2). Recall from Proposition 1 that $\mathcal{F}_k$ denotes the σ-algebra generated by all random directions drawn prior to iteration k.

Lemma 2 (Subspace gradient capture) *Suppose $p \geq 2$, and let $\boldsymbol{v}_k := \nabla f(\boldsymbol{x}_k)$. For the hybrid frame $\boldsymbol{D}_k$ constructed via* (1)–(2)*, it holds conditionally on $\mathcal{F}_k$ that*

$$\sum_{i=2}^{p} \left(\boldsymbol{v}_k^\top \boldsymbol{d}_i^{(k)}\right)^2 \overset{d}{=} \frac{\|\boldsymbol{v}_k\|_2^2}{n}\chi_{p-1}^2, \tag{16}$$

and the conditional expectation of the squared subspace gradient norm satisfies

$$\mathbb{E}\big[\|\hat{\boldsymbol{g}}_k\|_2^2 \mid \mathcal{F}_k\big] = \big(\boldsymbol{v}_k^\top \boldsymbol{d}_1^{(k)}\big)^2 \mathbf{1}_{\{\|\boldsymbol{m}_k\|_2 > 0\}} + \frac{p - \mathbf{1}_{\{\|\boldsymbol{m}_k\|_2>0\}}}{n}\|\boldsymbol{v}_k\|_2^2 \geq \frac{p-1}{n}\|\boldsymbol{v}_k\|_2^2. \tag{17}$$

Moreover, for every $q \in (0,1)$, there exist constants $\alpha > 0$ and $R > 1$, depending only on (n, p, q), such that whenever $\boldsymbol{v}_k \neq \boldsymbol{0}$,

$$\mathbb{P}\left(\|\hat{\boldsymbol{g}}_k\|_2 \geq \alpha\|\boldsymbol{v}_k\|_2, \quad \max_{1\leq i\leq p}\|\boldsymbol{d}_i^{(k)}\|_2 \leq R \;\middle|\; \mathcal{F}_k\right) \geq q. \tag{18}$$

Proof Conditionally on $\mathcal{F}_k$, $\boldsymbol{v}_k$ is deterministic and $\boldsymbol{d}_i^{(k)} = \boldsymbol{z}_i/\sqrt{n}$ $(i = 2, \dots, p)$ are i.i.d. $\mathcal{N}(\mathbf{0}, \frac{1}{n}\boldsymbol{I}_n)$ vectors, so $\boldsymbol{v}_k^\top \boldsymbol{d}_i^{(k)} \sim \mathcal{N}(0, \|\boldsymbol{v}_k\|_2^2/n)$ independently for $i = 2, \dots, p$, which proves (16). Since $\boldsymbol{d}_1^{(k)}$ is $\mathcal{F}_k$-measurable when $\|\boldsymbol{m}_k\|_2 > 0$ and satisfies $\boldsymbol{v}_k^\top \boldsymbol{d}_1^{(k)} \sim \mathcal{N}(0, \|\boldsymbol{v}_k\|_2^2/n)$ when $\boldsymbol{m}_k = \mathbf{0}$, taking conditional expectations in $\|\hat{\boldsymbol{g}}_k\|_2^2 = \sum_{i=1}^p (\boldsymbol{v}_k^\top \boldsymbol{d}_i^{(k)})^2$ gives (17).

Next, fix $q \in (0, 1)$. Because $p \geq 2$, χ_{p-1}^2 is almost surely positive, so there exists $\tau > 0$ (depending only on p and q) such that $\mathbb{P}(\chi_{p-1}^2 \geq \tau) \geq 1 - \frac{1-q}{2}$. Setting $\alpha = \sqrt{\tau/n}$ and noting that $\|\hat{\boldsymbol{g}}_k\|_2^2 = \sum_{i=1}^p (\boldsymbol{v}_k^\top \boldsymbol{d}_i^{(k)})^2 \geq \sum_{i=2}^p (\boldsymbol{v}_k^\top \boldsymbol{d}_i^{(k)})^2$, (16) implies

$$\mathbb{P}\left(\|\hat{\boldsymbol{g}}_k\|_2 \geq \alpha \|\boldsymbol{v}_k\|_2 \mid \mathcal{F}_k\right) \geq 1 - \frac{1-q}{2}.$$

Similarly, $\|\boldsymbol{d}_1^{(k)}\|_2 = 1$ when $\|\boldsymbol{m}_k\|_2 > 0$ and $\|\boldsymbol{d}_i^{(k)}\|_2^2 \overset{d}{=} \chi_n^2/n$ for each Gaussian column, so choosing $R > 1$ large enough that $\mathbb{P}(\chi_n^2/n \leq R^2) \geq 1 - \frac{1-q}{2p}$ ensures $\mathbb{P}(\max_{1 \leq i \leq p} \|\boldsymbol{d}_i^{(k)}\|_2 \leq R \mid \mathcal{F}_k) \geq 1 - \frac{1-q}{2}$ in both cases ($\|\boldsymbol{m}_k\|_2 > 0$ and $\boldsymbol{m}_k = \mathbf{0}$). A union bound gives (18). □

Equation (17) highlights the complementary roles of the two subspace components: whenever the prior accepted displacement $\boldsymbol{d}_1^{(k)}$ retains non-zero alignment with $\nabla f(\boldsymbol{x}_k)$, it contributes an $\mathcal{O}(1)$ dimension-free term $(\boldsymbol{v}_k^\top \boldsymbol{d}_1^{(k)})^2$ to $\|\hat{\boldsymbol{g}}_k\|_2^2$, while the $p - 1$ isotropic Gaussian directions guarantee a worst-case floor of $\frac{p-1}{n}\|\boldsymbol{v}_k\|_2^2$ even when $\boldsymbol{v}_k^\top \boldsymbol{d}_1^{(k)} = 0$.

3.2 Global convergence

We first show that the squared trust-region radii are summable along every sample path. Let

$$\mathcal{S} := \{k \geq 0 : \rho_k \geq \eta \text{ and } \|\boldsymbol{g}_k\|_2 \geq \kappa \Delta_k\} \tag{19}$$

denote the set of expansion iterations, and set $H := f(\boldsymbol{x}_0) - f_{\text{low}} < \infty$.

Lemma 3 (Sample-path summability of squared radii) *Suppose Assumption 1 holds, $\Delta_{\min} = 0$, and Δ_k is updated via* (11)*. Then along every sample path,*

$$\sum_{k=0}^{\infty} \Delta_k^2 \leq C_\Delta := \frac{\Delta_0^2 + \gamma_2^2 H/(\eta\kappa)}{1 - \gamma_1^2} < \infty. \tag{20}$$

In particular, $\mathbb{E}\big[\sum_{k=0}^{\infty} \Delta_k^2\big] \leq C_\Delta < \infty$ and $\lim_{k\to\infty} \Delta_k = 0$ deterministically.

Proof Because $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k^+$ if and only if $f(\boldsymbol{x}_k^+) < f(\boldsymbol{x}_k)$ and $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k$ otherwise, the sequence $\{f(\boldsymbol{x}_k)\}$ is non-increasing along every realization and $\sum_{k=0}^{\infty} \big(f(\boldsymbol{x}_k) - f(\boldsymbol{x}_{k+1})\big) \leq H$. For every $k \in \mathcal{S}$, (19) implies

$$f(\boldsymbol{x}_k) - f(\boldsymbol{x}_k^+) = \rho_k \Delta_k \|\boldsymbol{g}_k\|_2 \geq \eta\kappa\Delta_k^2 > 0,$$

so $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k^+$ and $\sum_{k\in\mathcal{S}} \Delta_k^2 \le H/(\eta\kappa)$. Furthermore, (11) gives $\Delta_{k+1}^2 \le \gamma_1^2\Delta_k^2 + \gamma_2^2\Delta_k^2\mathbf{1}_{\{k\in\mathcal{S}\}}$ for all $k \ge 0$. Summing from $k = 0$ to N and adding $(1-\gamma_1^2)\Delta_0^2$ to both sides yields

$$(1-\gamma_1^2)\sum_{k=0}^{N}\Delta_k^2 + \Delta_{N+1}^2 \le \Delta_0^2 + \gamma_2^2\sum_{k\in\mathcal{S}}\Delta_k^2 \le \Delta_0^2 + \frac{\gamma_2^2 H}{\eta\kappa}.$$

Since $1-\gamma_1^2 > 0$, dividing through and letting $N \to \infty$ establishes (20), which implies $\mathbb{E}\big[\sum_{k=0}^{\infty}\Delta_k^2\big] \le C_\Delta$ and $\lim_{k\to\infty}\Delta_k = 0$. □

Theorem 1 (Subsequential convergence) *Suppose Assumption 1 holds, $2 \le p \le n$, and $0 < \Delta_0 \le \Delta_{\max}$. Under the subspace construction* (1)–(2)*, the acceptance rule* (9)*, and the safeguarded radius update* (11) *with $\Delta_{\min} = 0$ and $\kappa > 0$,*

$$\liminf_{k\to\infty}\|\nabla f(\boldsymbol{x}_k)\|_2 = 0 \qquad \textit{almost surely.} \tag{21}$$

Proof Suppose, to the contrary, that $\liminf_{k\to\infty}\|\nabla f(\boldsymbol{x}_k)\|_2 > 0$ with positive probability. Then there exist a rational $\varepsilon > 0$ and an integer $K_0 \ge 0$ such that the event

$$E_{\varepsilon,K_0} := \{\|\nabla f(\boldsymbol{x}_k)\|_2 \ge \varepsilon,\ \ \forall\, k \ge K_0\}$$

satisfies $\mathbb{P}(E_{\varepsilon,K_0}) > 0$. Fix a probability threshold

$$q > \max\left\{\frac{1}{2},\ \frac{\log(1/\gamma_1)}{\log(\gamma_2/\gamma_1)}\right\} \in (0,1), \tag{22}$$

and let $\alpha > 0$ and $R > 1$ be the constants given by Lemma 2. Define the good-frame event

$$G_k := \left\{\|\hat{\boldsymbol{g}}_k\|_2 \ge \alpha\|\nabla f(\boldsymbol{x}_k)\|_2,\quad \max_{1\le i\le p}\|\boldsymbol{d}_i^{(k)}\|_2 \le R\right\},$$

and set $I_k := \mathbf{1}_{G_k}$ when $\nabla f(\boldsymbol{x}_k) \ne \mathbf{0}$ and $I_k := 1$ when $\nabla f(\boldsymbol{x}_k) = \mathbf{0}$, so that $\mathbb{E}[I_k \mid \mathcal{F}_k] \ge q$ holds everywhere on the probability space for all $k \ge 0$. Since $M_k := I_k - \mathbb{E}[I_k \mid \mathcal{F}_k]$ is a bounded martingale difference sequence ($|M_k| \le 1$), the strong law of large numbers for martingale differences gives $\lim_{N\to\infty} N^{-1}\sum_{k=K_0}^{K_0+N-1} M_k = 0$ almost surely, and therefore

$$\liminf_{N\to\infty}\frac{1}{N}\sum_{k=K_0}^{K_0+N-1} I_k \ge q \qquad \text{almost surely.} \tag{23}$$

Now consider any realization in E_{ε,K_0}. For $k \ge K_0$, whenever G_k occurs ($I_k = 1$), we have $Q_k \le \sqrt{p}\,R^2$, $\beta_k^2 = \|\boldsymbol{D}_k\|_2^2 \le \|\boldsymbol{D}_k\|_F^2 \le pR^2$, and by Lemma 1,

$$\|\boldsymbol{g}_k\|_2 \ge \|\hat{\boldsymbol{g}}_k\|_2 - \|\boldsymbol{g}_k - \hat{\boldsymbol{g}}_k\|_2 \ge \alpha\varepsilon - \frac{L\sqrt{p}\,R^2}{2}\Delta_k.$$

Since $\Delta_k \to 0$ deterministically by Lemma 3, there exists a finite index $K_1 \geq K_0$ along this realization such that for all $k \geq K_1$, $\gamma_2 \Delta_k \leq \Delta_{\max}$ and simultaneously, whenever $I_k = 1$,

$$\|\boldsymbol{g}_k\|_2 \geq \frac{\alpha\varepsilon}{2} \geq \kappa\Delta_k \qquad \text{and} \qquad \rho_k \geq 1 - \frac{L(\sqrt{p}+p)R^2\Delta_k}{\alpha\varepsilon} \geq \eta,$$

where the second inequality uses Proposition 2. Thus, on E_{ε,K_0}, for all $k \geq K_1$, the update (11) gives $\Delta_{k+1} = \gamma_2\Delta_k$ whenever $I_k = 1$ and $\Delta_{k+1} \geq \gamma_1\Delta_k$ whenever $I_k = 0$, or equivalently

$$\log \Delta_{k+1} - \log \Delta_k \geq I_k \log \gamma_2 + (1 - I_k) \log \gamma_1, \qquad \forall\, k \geq K_1. \tag{24}$$

Averaging (24) and invoking (23) together with (22) shows that for almost every realization in E_{ε,K_0},

$$\liminf_{N\to\infty} \frac{\log \Delta_{K_0+N} - \log \Delta_{K_0}}{N} \geq q \log \gamma_2 + (1-q) \log \gamma_1 > 0.$$

Consequently, $\Delta_k \to \infty$, contradicting $\Delta_k \to 0$ from Lemma 3, and therefore $\mathbb{P}(E_{\varepsilon,K_0}) = 0$, which proves (21). □

To upgrade Theorem 1 from $\liminf_{k\to\infty} \|\nabla f(\boldsymbol{x}_k)\|_2 = 0$ to full convergence $\lim_{k\to\infty} \|\nabla f(\boldsymbol{x}_k)\|_2 = 0$ without imposing external renewal assumptions on unsuccessful iterations [2, 13], we must control how far the iterate $\boldsymbol{x}_k$ can travel across the index set of large gradients,

$$\mathcal{K}_\varepsilon := \{k \geq 0 : \|\nabla f(\boldsymbol{x}_k)\|_2 > \varepsilon\}, \qquad \varepsilon > 0.$$

Although a single iteration $k \in \mathcal{K}_\varepsilon$ may draw an unfavorable frame ($I_k = 0$) and fail to reduce f, the indicator $\mathbf{1}_{\{k\in\mathcal{K}_\varepsilon\}}$ and radius Δ_k are $\mathcal{F}_k$-measurable prior to drawing $\boldsymbol{D}_k$, whereas a good frame G_k occurs with conditional probability at least $1/2$. Taking conditional expectations therefore transfers summability from the good-frame steps to all of $\mathcal{K}_\varepsilon$, while Proposition 1 bounds the random spectral factor $\beta_k = \|\boldsymbol{D}_k\|_2$.

Lemma 4 (Summability along large-gradient iterations) *Under the assumptions of Theorem 1, for every $\varepsilon > 0$,*

$$\mathbb{E}\left[\sum_{k\in\mathcal{K}_\varepsilon} \Delta_k\right] < \infty \qquad \textit{and} \qquad \mathbb{E}\left[\sum_{k\in\mathcal{K}_\varepsilon} \beta_k\Delta_k\right] < \infty. \tag{25}$$

Consequently, $\sum_{k\in\mathcal{K}_\varepsilon} \Delta_k < \infty$ and $\sum_{k\in\mathcal{K}_\varepsilon} \beta_k\Delta_k < \infty$ almost surely.

Proof Fix $\varepsilon > 0$ and $q = 1/2$, and let $\alpha > 0$, $R > 1$, and G_k be given by Lemma 2 so that $\mathbb{P}(G_k \mid \mathcal{F}_k) \geq 1/2$ on $\{k \in \mathcal{K}_\varepsilon\}$. Define the deterministic threshold

$$\delta_\varepsilon := \min\left\{\frac{\alpha\varepsilon}{L\sqrt{p}\,R^2}, \frac{(1-\eta)\alpha\varepsilon}{L(\sqrt{p}+p)R^2}\right\} > 0.$$

Whenever $k \in \mathcal{K}_\varepsilon$, G_k holds, and $\Delta_k \le \delta_\varepsilon$, Lemma 1 and Proposition 2 imply $\|\boldsymbol{g}_k\|_2 \ge \alpha\varepsilon/2 > 0$ and $\rho_k \ge \eta > 0$, so $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k^+$ and

$$f(\boldsymbol{x}_k) - f(\boldsymbol{x}_{k+1}) = \rho_k \Delta_k \|\boldsymbol{g}_k\|_2 \ge c_\varepsilon \Delta_k, \qquad c_\varepsilon := \frac{\eta\alpha\varepsilon}{2} > 0.$$

Conversely, whenever $\Delta_k > \delta_\varepsilon$, we have $\Delta_k < \Delta_k^2/\delta_\varepsilon$. Combining both cases gives, for every $k \ge 0$,

$$\Delta_k \mathbf{1}_{\{k\in\mathcal{K}_\varepsilon\}} \mathbf{1}_{G_k} \le \frac{f(\boldsymbol{x}_k) - f(\boldsymbol{x}_{k+1})}{c_\varepsilon} + \frac{\Delta_k^2}{\delta_\varepsilon}.$$

Summing over $k \ge 0$ and applying Lemma 3 yields the deterministic sample-path bound

$$\sum_{k=0}^{\infty} \Delta_k \mathbf{1}_{\{k\in\mathcal{K}_\varepsilon\}} \mathbf{1}_{G_k} \le \frac{H}{c_\varepsilon} + \frac{C_\Delta}{\delta_\varepsilon} < \infty.$$

Let $a_k := \Delta_k \mathbf{1}_{\{k\in\mathcal{K}_\varepsilon\}} \ge 0$. Since a_k is $\mathcal{F}_k$-measurable and $\mathbb{E}[\mathbf{1}_{G_k} \mid \mathcal{F}_k] \ge 1/2$ whenever $a_k > 0$, Tonelli's theorem and the tower property give

$$\begin{aligned}\frac{1}{2}\,\mathbb{E}\left[\sum_{k\in\mathcal{K}_\varepsilon} \Delta_k\right] &= \sum_{k=0}^{\infty} \frac{1}{2}\mathbb{E}[a_k] \le \sum_{k=0}^{\infty} \mathbb{E}\big[a_k \mathbb{E}[\mathbf{1}_{G_k} \mid \mathcal{F}_k]\big] \\ &= \mathbb{E}\left[\sum_{k=0}^{\infty} a_k \mathbf{1}_{G_k}\right] \le \frac{H}{c_\varepsilon} + \frac{C_\Delta}{\delta_\varepsilon} < \infty,\end{aligned}$$

proving the first inequality in (25).

Next, by Proposition 1 (3), $\mathbb{E}[\|\boldsymbol{d}_i^{(k)}\|_2^2 \mid \mathcal{F}_k] = 1$ for all $i = 1, \dots, p$, so

$$\mathbb{E}[\beta_k^2 \mid \mathcal{F}_k] \le \mathbb{E}\big[\|\boldsymbol{D}_k\|_{\mathrm{F}}^2 \,\big|\, \mathcal{F}_k\big] = \sum_{i=1}^{p} \mathbb{E}\big[\|\boldsymbol{d}_i^{(k)}\|_2^2 \,\big|\, \mathcal{F}_k\big] = p. \tag{26}$$

By conditional Jensen's inequality, $\mathbb{E}[\beta_k \mid \mathcal{F}_k] \le (\mathbb{E}[\beta_k^2 \mid \mathcal{F}_k])^{1/2} \le \sqrt{p}$. Because $a_k \ge 0$ is $\mathcal{F}_k$-measurable, Tonelli's theorem again yields

$$\mathbb{E}\left[\sum_{k\in\mathcal{K}_\varepsilon} \beta_k \Delta_k\right] = \sum_{k=0}^{\infty} \mathbb{E}\big[a_k \mathbb{E}[\beta_k \mid \mathcal{F}_k]\big] \le \sqrt{p}\,\mathbb{E}\left[\sum_{k\in\mathcal{K}_\varepsilon} \Delta_k\right] < \infty,$$

establishing (25) and hence almost-sure finiteness. □

Combining (26) with Lemma 3 also yields $\sum_{k=0}^{\infty} \mathbb{E}[\beta_k^2 \Delta_k^2] \le pC_\Delta < \infty$, so $\sum_{k=0}^{\infty} \beta_k^2 \Delta_k^2 < \infty$ almost surely and

$$\lim_{k\to\infty} \beta_k \Delta_k = 0 \qquad \text{almost surely.} \tag{27}$$

Theorem 2 (Full global convergence) *Under the assumptions of Theorem 1,*

$$\lim_{k\to\infty} \|\nabla f(\boldsymbol{x}_k)\|_2 = 0 \qquad \textit{almost surely.} \tag{28}$$

Proof By Theorem 1, Lemma 4, and (27), the event

$$\Omega_0 := \left\{ \liminf_{k\to\infty} \|\nabla f(\boldsymbol{x}_k)\|_2 = 0,\ \lim_{k\to\infty} \beta_k \Delta_k = 0 \right\} \cap \bigcap_{\varepsilon\in\mathbb{Q}_{>0}} \left\{ \sum_{k\in\mathcal{K}_\varepsilon} \beta_k \Delta_k < \infty \right\}$$

is a countable intersection of almost-sure events and satisfies $\mathbb{P}(\Omega_0) = 1$. Suppose, for contradiction, that on some realization in Ω_0 the sequence $\|\nabla f(\boldsymbol{x}_k)\|_2$ does not converge to zero. Since $\liminf_{k\to\infty} \|\nabla f(\boldsymbol{x}_k)\|_2 = 0$, there exist a rational $\varepsilon \in \mathbb{Q}_{>0}$ and infinitely many disjoint index intervals $[a_j, b_j]$ $(a_j < b_j)$ such that

$$\|\nabla f(\boldsymbol{x}_{a_j})\|_2 \le \varepsilon, \quad \|\nabla f(\boldsymbol{x}_{b_j})\|_2 > 2\varepsilon, \quad \|\nabla f(\boldsymbol{x}_k)\|_2 > \varepsilon \ \ (\forall\, a_j < k < b_j).$$

Because $\|\boldsymbol{x}_{k+1} - \boldsymbol{x}_k\|_2 \le \|\boldsymbol{D}_k \boldsymbol{s}_s\|_2 \le \beta_k \Delta_k$ for all $k \ge 0$, Assumption 1 and the triangle inequality give

$$\varepsilon < \|\nabla f(\boldsymbol{x}_{b_j})\|_2 - \|\nabla f(\boldsymbol{x}_{a_j})\|_2 \le L \left(\beta_{a_j} \Delta_{a_j} + \sum_{k=a_j+1}^{b_j-1} \beta_k \Delta_k \right).$$

On Ω_0, as $j \to \infty$, we have $\beta_{a_j} \Delta_{a_j} \to 0$, and since $\{a_j + 1, \dots, b_j - 1\} \subset \mathcal{K}_\varepsilon$ with $\sum_{k\in\mathcal{K}_\varepsilon} \beta_k \Delta_k < \infty$, the tail sum $\sum_{k=a_j+1}^{b_j-1} \beta_k \Delta_k \to 0$. The right-hand side therefore converges to zero as $j \to \infty$, contradicting $\varepsilon > 0$. Thus (28) holds everywhere on Ω_0, and hence almost surely. □

4 Numerical Experiments

We evaluate `MpSub` in two settings. Section 4.1 compares `MpSub` with the zeroth-order baseline MeZO [20] on full-parameter language model fine-tuning under matched forward-pass budgets, and examines sensitivity to the initial radius Δ_0 and the subspace dimension p. Section 4.2 then evaluates the effect of each design choice in `MpSub` on a 1,000-dimensional smooth nonquadratic test problem.

4.1 Language model fine-tuning

Language model experiments are conducted on CommitmentBank (CB) from the SuperGLUE benchmark [25], a three-class natural language inference task. Following [20], we sample 100 training examples, reserve 50 examples for development, and evaluate on the standard 56-example validation split as the test set (where one test example corresponds to $1/56 \approx 0.018$ in accuracy). We fine-tune two autoregressive models from the OPT family [31]: OPT-125M ($n \approx 1.25 \times 10^8$ parameters) and OPT-350M ($n \approx 3.5 \times 10^8$ parameters). All parameters are stored and optimized in FP32 with a minibatch size of $|\mathcal{B}_k| = 8$. Results are reported as means and min–max ranges over three random seeds.

Because zeroth-order methods do not compute gradients, computational cost is measured by the total number of forward passes on the training objective. Each run is allotted a budget of 8,400 training forward passes (excluding periodic development evaluations). For `MpSub` with $p = 20$, each iteration uses $2p+2 = 42$ forward passes on a fixed minibatch $\mathcal{B}_k$, corresponding to 200 trust-region iterations per run. MeZO uses 2 forward passes per step and resamples a minibatch at every step, executing 4,200 steps under the same 8,400-evaluation budget. Across both models, `MpSub` uses a single fixed configuration: $p = 20$, $\Delta_0 = 10^{-1}$, $\gamma_1 = 0.5$, $\gamma_2 = 2.0$, $\eta = 0.1$, $\Delta_{\min} = 10^{-12}$, and $\Delta_{\max} = 1.0$. For MeZO, the perturbation scale is fixed at $\epsilon = 10^{-3}$, while the learning rate η_{M} is tuned by mean development accuracy over $\{10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}\}$ on OPT-125M and $\{10^{-7}, 10^{-6}, 10^{-5}\}$ on OPT-350M, requiring 5× and 3× the single-run evaluation budget, respectively.

Table 1 Matched-budget fine-tuning results on CommitmentBank (8,400 training forward passes per run). Values are means over three random seeds, with min–max ranges in parentheses. The MeZO learning rate η_{M} is selected by mean development accuracy over the candidate grid.

Model	Method	Dev loss	Dev acc.	Test acc.
OPT-125M	`MpSub` ($\Delta_0 = 10^{-1}$)	0.674 (0.620–0.723)	0.773 (0.740–0.820)	0.673 (0.661–0.679)
	MeZO ($\eta_{\mathrm{M}} = 10^{-6}$)	0.703 (0.625–0.784)	0.760 (0.740–0.800)	0.685 (0.679–0.696)
OPT-350M	`MpSub` ($\Delta_0 = 10^{-1}$)	0.738 (0.715–0.774)	0.787 (0.760–0.820)	0.690 (0.679–0.714)
	MeZO ($\eta_{\mathrm{M}} = 10^{-6}$)	0.679 (0.582–0.743)	0.780 (0.720–0.880)	0.685 (0.679–0.696)

Table 1 and Figure 1 compare the two methods under the 8,400-evaluation budget. On OPT-125M, default `MpSub` attains a lower mean development loss (0.674 vs. 0.703) and higher development accuracy (0.773 vs. 0.760) than grid-searched MeZO, with mean test accuracies of 0.673 versus 0.685. On OPT-350M, MeZO reaches a lower cross-entropy loss mid-training, whereas `MpSub` exhibits smaller seed-to-seed variance and achieves a mean test accuracy of 0.690 and development accuracy of 0.787, compared to 0.685 and 0.780 for tuned MeZO. On both models, the difference in mean test accuracy between untuned `MpSub` and grid-searched MeZO is within a single test example (0.012 on OPT-125M and 0.006 on OPT-350M) with overlapping seed ranges.

Figure 2 compares the sensitivity of MeZO to its learning rate $\eta_{\mathrm{M}} \in [10^{-8}, 10^{-4}]$ with the sensitivity of `MpSub` to its initial radius $\Delta_0 \in \{10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}, 3 \times 10^{-1}\}$ on OPT-125M. Because MeZO holds η_{M} fixed throughout training, shifting η_{M} by one order of magnitude from 10^{-6} reduces mean test accuracy by 0.048 at 10^{-7} and by 0.084 at 10^{-5}. In contrast, `MpSub` maintains a mean test accuracy between 0.655 and 0.679 across the 300-fold range $\Delta_0 \in [10^{-3}, 3 \times 10^{-1}]$. Because Δ_0 serves only as the initial value for the geometric update (10), a small choice such as $\Delta_0 = 10^{-3}$ doubles on successful steps and reaches $\Delta_k \approx 10^{-1}$ within $\lceil \log_2(10^2) \rceil = 7$ iterations. Performance degrades only when $\Delta_0 \geq 1.0$ (mean test accuracies of 0.554, 0.488, and 0.464

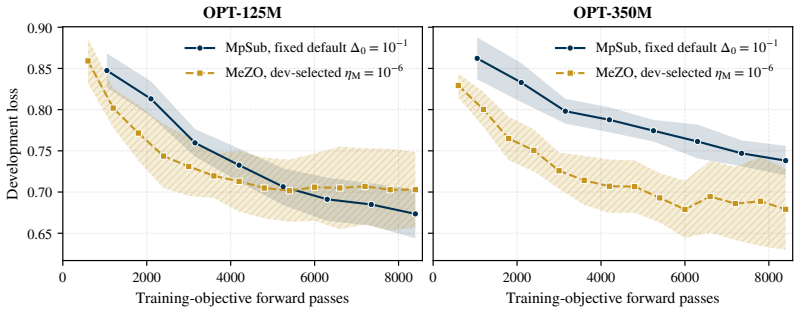


Fig. 1 Development loss versus training forward passes for OPT-125M (left) and OPT-350M (right), evaluated every 25 steps (1,050 forward passes) for `MpSub` and every 300 steps (600 forward passes) for MeZO. Curves show three-seed means; shaded bands indicate ±1 standard error of the mean.

at $\Delta_0 \in \{1.0, 3.0, 10.0\}$), where the initial central-difference perturbations (4) incur large truncation error (Lemma 1) and propose overly large initial steps.

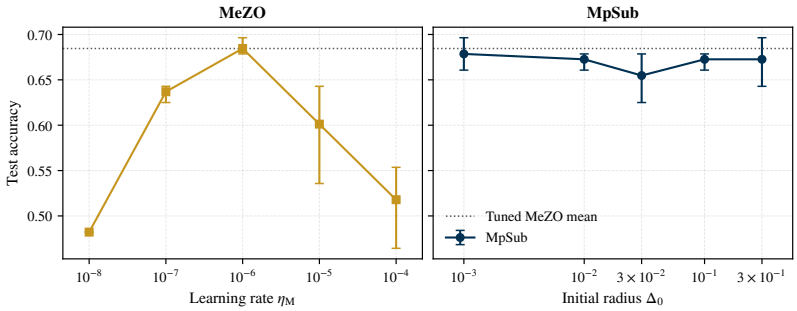


Fig. 2 Step-size sensitivity on OPT-125M under 8,400 forward passes. Markers show three-seed mean test accuracy and vertical bars indicate min–max ranges. Left: MeZO across learning rates η_M. Right: `MpSub` across initial radii Δ_0; the horizontal dotted line marks the tuned MeZO mean (0.685).

We next examine the effect of the subspace dimension $p \in \{5, 10, 15, 20, 25, 30\}$ on OPT-125M in Table 2 under two protocols: (i) a fixed horizon of 100 iterations (single seed) to measure per-iteration progress and wall-clock time, and (ii) a matched budget of at most 8,400 forward passes across three seeds, under which `MpSub` executes $\lfloor 8{,}400/(2p{+}2) \rfloor$ iterations. At a fixed 100-iteration horizon, increasing p from 5 to 20 raises development accuracy from 0.700 to 0.780 and test accuracy from 0.643 to 0.679, consistent with the $(p-1)/n$ scaling in Lemma 2, while per-step wall-clock time grows linearly with $2p+2$ from 3.0 s to 10.1 s. Under the fixed 8,400-evaluation budget, smaller values of p capture a smaller fraction of the gradient energy per iteration but execute more iterations (700 iterations at $p = 5$ vs. 200 at $p = 20$); consequently, mean

test accuracy remains within 0.667–0.685 across all p, while $p \geq 20$ achieves higher mean development accuracy (0.773–0.780 vs. 0.740–0.760 for $p \leq 15$).

Table 2 Scaling with subspace dimension p for OPT-125M on CommitmentBank ($\Delta_0 = 10^{-1}$). Left: fixed horizon of 100 steps (single seed, wall-clock time per step on an NVIDIA RTX 4060 Ti GPU). Right: matched budget of at most 8,400 forward passes ($\lfloor 8{,}400/(2p{+}2) \rfloor$ steps; means and min–max ranges over three seeds).

		Fixed horizon (100 steps)			Matched budget ($\leq$ 8,400 passes, 3 seeds)			
p	Fwd/step	Dev acc.	Test acc.	Time/step	Steps	Dev loss	Dev acc.	Test acc.
5	12	0.700	0.643	3.0 s	700	0.670 (0.592–0.743)	0.760 (0.720–0.820)	0.685 (0.679–0.696)
10	22	0.720	0.661	4.4 s	381	0.683 (0.592–0.776)	0.740 (0.680–0.800)	0.679 (0.679–0.679)
15	32	0.740	0.661	6.0 s	262	0.673 (0.607–0.751)	0.747 (0.720–0.780)	0.667 (0.661–0.679)
20	42	0.780	0.679	7.7 s	200	0.674 (0.620–0.723)	0.773 (0.740–0.820)	0.673 (0.661–0.679)
25	52	0.780	0.679	8.7 s	161	0.676 (0.610–0.738)	0.773 (0.740–0.820)	0.673 (0.661–0.679)
30	62	0.760	0.661	10.1 s	135	0.682 (0.609–0.740)	0.780 (0.760–0.820)	0.679 (0.679–0.679)

4.2 Component ablation

To evaluate `MpSub` in the smooth deterministic setting of Section 3, we minimize a 1,000-dimensional nonquadratic function with rotated coordinates and power-law spectrum:

$$f(\boldsymbol{x}) = \frac{1}{2}\sum_{j=1}^{n} \lambda_j z_j^2 + c_4 \sum_{j=1}^{n} \lambda_j^2 z_j^4, \qquad \boldsymbol{z} = (\boldsymbol{I}_n - 2\boldsymbol{v}\boldsymbol{v}^\top)(\boldsymbol{x} - \boldsymbol{x}_\star), \tag{29}$$

where $n = 1{,}000$, $\lambda_j = 50j^{-3/4}$ ($\lambda_1/\lambda_n \approx 280$), $c_4 = 0.05$, $\boldsymbol{v} \in \mathbb{R}^n$ is a fixed unit Householder vector, and $\boldsymbol{x}_0 = \boldsymbol{0}$. The optimizer accesses only function values $f(\boldsymbol{x})$, while exact gradients $\nabla f(\boldsymbol{x}_k)$ are evaluated offline. Table 3 reports the relative objective $f(\boldsymbol{x}_K)/f(\boldsymbol{x}_0)$, relative gradient norm $\|\nabla f(\boldsymbol{x}_K)\|_2/\|\nabla f(\boldsymbol{x}_0)\|_2$, and step acceptance rate after 8,400 function evaluations across five random seeds.

In the upper part of Table 3, we modify one component of `MpSub` at a time. Adding the expansion condition $\|\boldsymbol{g}_k\|_2 \geq \kappa\Delta_k$ ($\kappa = 10^{-3}$) from (11) produces the exact same iterates as Algorithm 1 ($\kappa = 0$), because $\|\boldsymbol{g}_k\|_2 \geq \kappa\Delta_k$ holds at every successful step during the 200 iterations. Replacing the momentum direction $\boldsymbol{d}_1^{(k)}$ with an independent Gaussian vector increases the final relative objective from 0.158 to 0.186 and the relative gradient norm from 0.194 to 0.223 with non-overlapping seed ranges (0.150–0.170 vs. 0.168–0.204); as shown in (17), retaining $\boldsymbol{d}_1^{(k)}$ contributes the additional term $(\boldsymbol{v}_k^\top \boldsymbol{d}_1^{(k)})^2$ to

Table 3 Component ablation on the smooth deterministic benchmark (29) (n = 1,000, 8,400 function evaluations; means and min–max ranges over five random seeds).

Variant	Steps	$f(\boldsymbol{x}_K)/f(\boldsymbol{x}_0)$	$\|\nabla f(\boldsymbol{x}_K)\|_2/\|\nabla f(\boldsymbol{x}_0)\|_2$	Accept
`MpSub` ($p = 20, \Delta_0 = 10^{-1}$)	200	0.158 (0.150–0.170)	0.194 (0.184–0.219)	0.58
`MpSub` + safeguard ($\kappa = 10^{-3}$)	200	0.158 (0.150–0.170)	0.194 (0.184–0.219)	0.58
No momentum ($p = 20$ random)	200	0.186 (0.168–0.204)	0.223 (0.211–0.231)	0.58
Classical acceptance ($\rho_k \geq \eta$)	200	0.188 (0.177–0.213)	0.222 (0.209–0.267)	0.50
`MpSub` ($p = 2$ linear model)	1,400	0.320 (0.296–0.330)	0.288 (0.276–0.294)	0.54
`MpSub` from small $\Delta_0 = 10^{-3}$	200	0.174 (0.164–0.192)	0.207 (0.194–0.223)	0.58
`MpSub` from large $\Delta_0 = 1.0$	200	0.170 (0.158–0.180)	0.205 (0.189–0.216)	0.57
Fixed radius ($\Delta_k \equiv 10^{-3}$)	200	0.667 (0.664–0.669)	0.517 (0.511–0.520)	1.00
Fixed radius ($\Delta_k \equiv 1.0$)	200	1.000 (1.000–1.000)	1.000 (1.000–1.000)	0.00

the expected squared norm of the projected gradient whenever the gradient correlates with the previous displacement. Reducing the subspace dimension from $p = 20$ to $p = 2$ (1,400 steps at 6 evaluations per step) more than doubles the final relative objective (0.320 vs. 0.158), while replacing the acceptance condition $\rho_k > 0$ in (9) with the threshold $\rho_k \geq \eta$ discards descending trial steps with $0 < \rho_k < \eta$, reducing the acceptance rate from 0.58 to 0.50 and increasing the final relative objective to 0.188.

In the lower part of Table 3, we compare the adaptive radius update (10) with fixed trust-region radii. When Δ_k is held constant, $\Delta_k \equiv 10^{-3}$ accepts every step but makes slow progress ($f(\boldsymbol{x}_K)/f(\boldsymbol{x}_0) = 0.667$) because the step norm is restricted to 10^{-3}, whereas $\Delta_k \equiv 1.0$ overshoots along the high-curvature directions of (29) and rejects every trial point (1.000). When Δ_k is updated via (10), `MpSub` initialized at either $\Delta_0 = 10^{-3}$ or $\Delta_0 = 1.0$ adjusts Δ_k within the first few iterations and reaches relative objectives of 0.174 and 0.170, close to the result obtained from $\Delta_0 = 10^{-1}$ (0.158).

5 Conclusion

We have presented `MpSub`, a momentum p-dimensional subspace trust-region method for high-dimensional derivative-free optimization and full-parameter language model fine-tuning. By combining the normalized displacement of the most recent accepted step with $p - 1$ unorthogonalized Gaussian directions, `MpSub` relies on the near-orthonormality of high-dimensional Gaussian frames (Proposition 1) to regenerate exploratory directions in place from scalar seeds, requiring only $\mathcal{O}(n)$ auxiliary memory and $2p + 2$ function evaluations per iteration. For smooth deterministic objectives, we derived finite-difference error and single-step descent bounds for general non-orthogonal frames and proved unconditional almost-sure global convergence ($\lim_{k\to\infty} \|\nabla f(\boldsymbol{x}_k)\|_2 = 0$) under a safeguarded radius update. Numerical experiments on a 1,000-dimensional nonquadratic test problem and on SuperGLUE CommitmentBank fine-tuning of OPT-125M and OPT-350M demonstrated the practical contribution of each

algorithmic component and showed that trust-region radius adaptation enables `MpSub` to match grid-searched MeZO using a single default configuration.

Several extensions merit further investigation. First, while our convergence analysis treats a deterministic objective f, language model fine-tuning evaluates a sampled minibatch objective f_k at each iteration; extending the analysis of Section 3 to stochastic minibatch evaluations under bounded variance is an important next step. Second, within each p-dimensional subspace $\mathcal{S}_k$, incorporating low-cost secant or diagonal curvature estimates—and parallelizing the $2p$ independent directional forward passes across devices—could further accelerate convergence while preserving the $\mathcal{O}(p)$ evaluation cost and $\mathcal{O}(n)$ memory footprint.

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Statements and Declarations

The authors have no competing interests or funding to declare. The dataset analyzed during the current study is available in the SuperGLUE benchmark repository (`https://super.gluebenchmark.com`), and the source code for `MpSub` and the experiment scripts are available at `https://github.com/Julius-Valerius/MpSub`.

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